Introduction to Basic terminology

Module II

Basic terminology

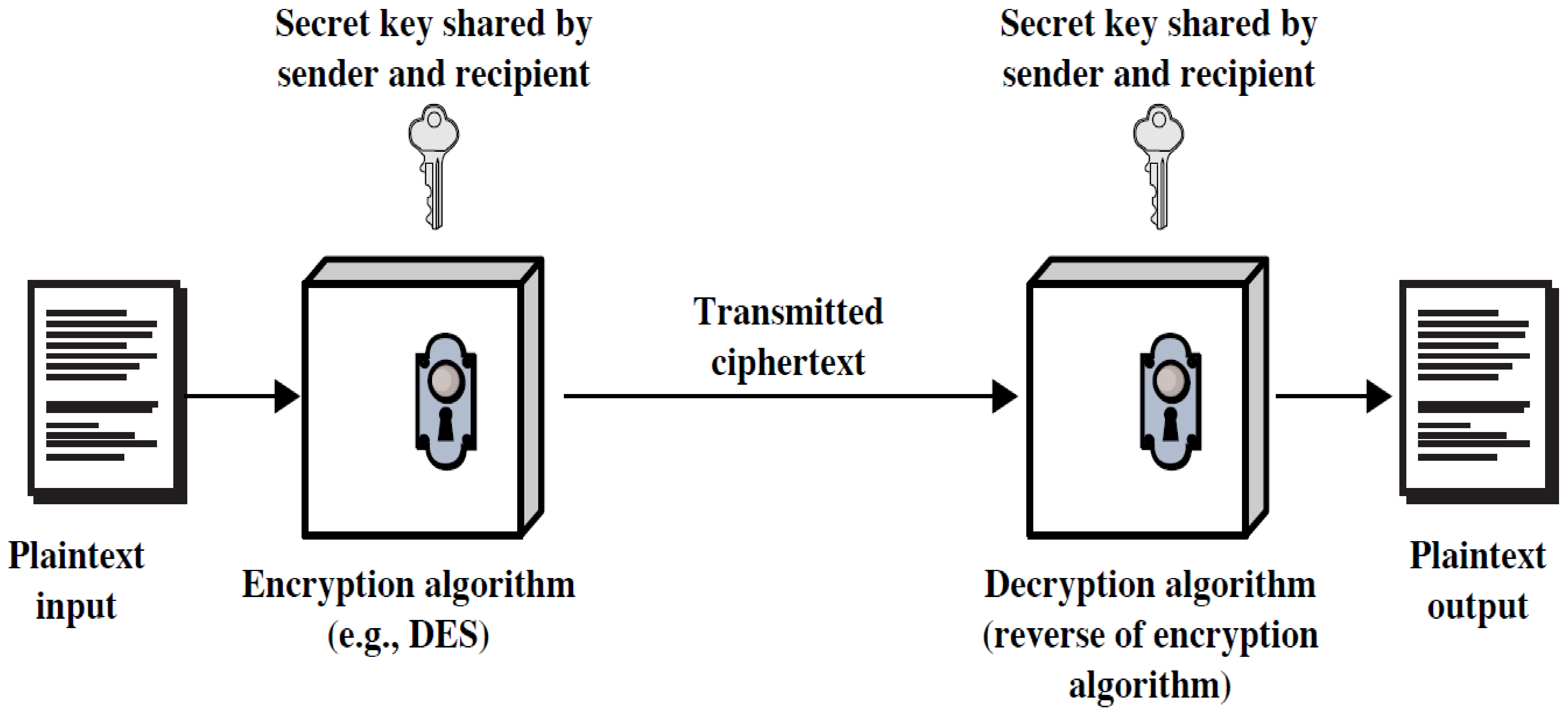
- Plaintext: original message to be encrypted

- Ciphertext: the encrypted message

- Enciphering or encryption: the process of converting plaintext into ciphertext

- Encryption algorithm: performs encryption
  - Two inputs: a plaintext and a secret key

Symmetric Cipher Model



Basic terminology cntd..

- Deciphering or decryption: recovering plaintext from ciphertext
- Decryption algorithm: performs decryption
  - Two inputs: ciphertext and secret key
- Secret key: same key used for encryption and decryption
  - Also referred to as a symmetric key

Basic terminology cntd..

- Cipher or cryptographic system : a scheme for encryption and decryption

- Cryptography: science of studying ciphers

- Cryptanalysis: science of studying attacks against cryptographic systems

- Cryptology: cryptography + cryptanalysis

Ciphers

- Symmetric cipher: same key used for encryption and decryption
  - Block cipher: encrypts a block of plaintext at a time (typically 64 or 128 bits)
  - Stream cipher: encrypts data one bit or one byte at a time
- Asymmetric cipher: different keys used for encryption and decryption

Symmetric Encryption

- or conventional / secret-key / single-key
- sender and recipient share a common key
- all classical encryption algorithms are symmetric
- The only type of ciphers prior to the invention of asymmetric-key ciphers in 1970's
- by far most widely used

Symmetric Encryption cntd..

- Mathematically:

  $Y = E_K(X)$ or $Y = E(K, X)$

  $X = D_K(Y)$ or $X = D(K, Y)$
- X = plaintext
- Y = ciphertext
- K = secret key
- E = encryption algorithm
- D = decryption algorithm
- Both E and D are known to public

Cryptanalysis

- Objective: to recover the plaintext of a ciphertext or, more typically, to recover the secret key.
- Kerkhoff's principle: the adversary knows all details about a cryptosystem except the secret key.
- Two general approaches:
  - brute-force attack
  - non-brute-force attack (cryptanalytic attack)

Brute-Force Attack

- Try every key to decipher the ciphertext.
- On average, need to try half of all possible keys
- Time needed proportional to size of key space

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/µs	Time required at 10^6 decryptions/µs
32	$2^{32} = 4.3 \times 10^9$	2^{31} µs = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	2^{55} µs = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	2^{127} µs = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	2^{167} µs = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	2×10^{26} µs = 6.4×10^{12} years	6.4×10^6 years

Cryptanalytic Attacks

- There are four common types of cryptanalysis attacks. They are,
  - Ciphertext-only attack
  - Known-plaintext attack
  - Chosen-plaintext attack
  - Chosen-ciphertext attack

Cryptography is the science and art of creating secret codes.

Cryptanalysis is the science and art of breaking those codes. This is needed, not to break other people's code, but to learn how vulnerable our cryptosystem is. The study of cryptanalysis helps us create better secret codes.

Cipher text only attack

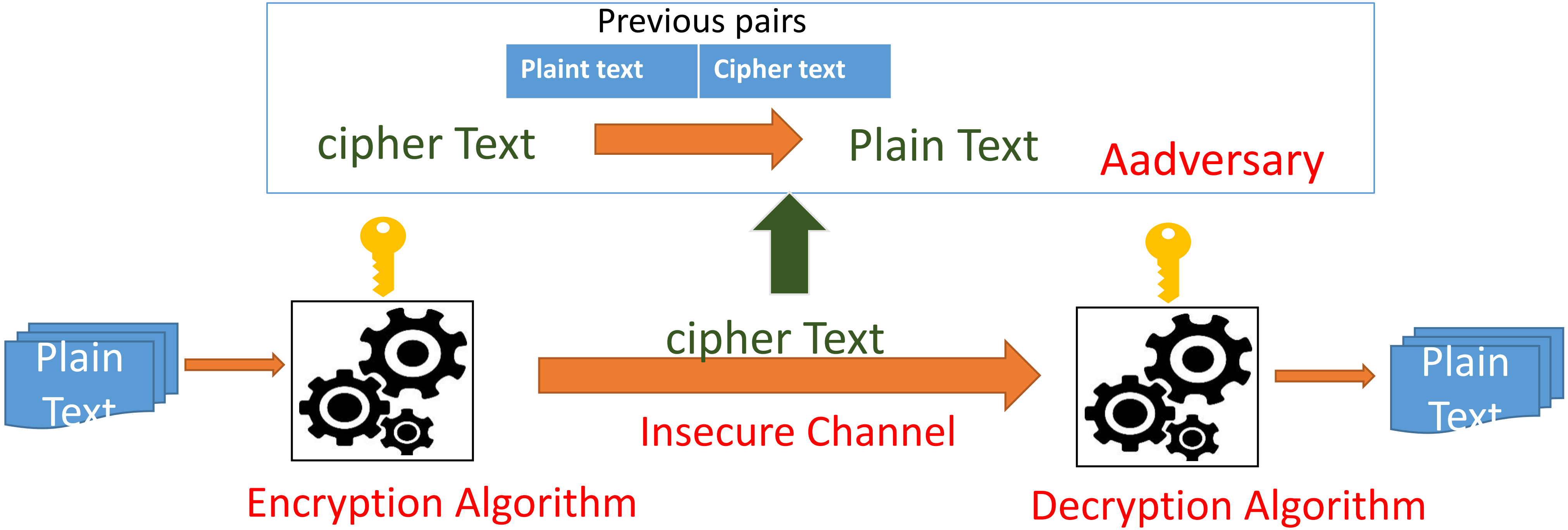
- In this, Intruder has access to only some ciphertext and tires to find the corresponding key and the plaintext. To thwart the decryption of a message by an adversary, a cipher must be very resisting to this type of attack.

- Statistical attack: The cryptanalyst can benefit some Inherent characteristics of the plaintext language to launch a statistical attack. For example letter E is the most-frequently used letter in English text.

- Pattern attack: Some ciphers may hide the charcteristics of the language, but may create some patterns in the ciphertext. An adversary can exploit and use pattern attack to break the cipher. Therefore, it is Important to use ciphers that make the ciphertext look as random as possible.

Known-Plaintext attack

- In this, Intruder has access to some plaintext/ciphertext text pairs in addition to the Interpreted cipher text.



Chosen Plaintext attack

- The chosen plaintext attack is similar to the known-plaintext attack, but the plaintext/ciphertext pairs have been chosen by the attacker. It can happen it adversary has access to sender’s computer.

Chosen Ciphertext attack

- The chosen ciphertext attack is similar to the chosen-plaintext attack, except that adversary chooses some ciphertext and decrypts it to form a ciphertext/plaintext pair. This can happen if adversary has access to receiver’s computer.